

DANIEL MARÍN PINA

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RESEARCH ACTIVITIES

I am a theoretical astrophysicist based in the University of Heidelberg, Germany. My main interest is in stellar and black hole dynamics in star clusters, and the formation of dormant star-black hole binaries and gravitational waves from binary black hole mergers. My goal for the next years is to develop models for the population of these dormant star-black hole binaries, in preparation for the next *Gaia* data release and results from other massive surveys. Furthermore, I want to keep working on modelling the dynamical formation of binary black hole mergers in star clusters. This will help us understand the production of eccentric gravitational waves and the seeding and growth of intermediate-mass black holes.

EDUCATION

- PhD Cum Laude in Astronomy and Astrophysics** Barcelona, Spain
Institute of Cosmos Sciences, University of Barcelona 2021 - 2025
- Thesis title: Black hole dynamics in star clusters: Eccentric mergers and star-black hole binaries
 - Grade: Excellent Cum Laude (maximum possible grade)
 - Advisor: Prof. Mark Gieles
- Master in Astrophysics, Particle Physics and Cosmology** Barcelona, Spain
University of Barcelona 2019 - 2020
- Master's thesis title: Event horizon in a compact binary coalescence
 - Advisor: Prof. Roberto Emparan
- Bachelor in Physics** Barcelona, Spain
University of Barcelona 2015 - 2019
NTNU Norwegian University of Science and Technology 2018

WORK EXPERIENCE

- DFG Postdoctoral Fellow** Heidelberg, Germany
Institute for Theoretical Astrophysics, University of Heidelberg 2025 - 2028 (ongoing)
- FPI Predoctoral Fellow** Barcelona, Spain
Institute of Cosmos Sciences, University of Barcelona 2021 - 2025
- Gaia DPAC Postgraduate Researcher** Barcelona, Spain
Technological Unit, Institute of Cosmos Sciences, University of Barcelona 2019 - 2021

AWARDS AND HONORS

- **PhD Thesis Cum Laude** University of Barcelona
“Black hole dynamics in star clusters: Eccentric mergers and star-black hole binaries” 2025
- **Best Poster Presentation** ESO Garching, Germany
“Two in a million: The interplay between binaries and star clusters” 2023
- **Best Master's Thesis** University of Barcelona
“Event horizon in a compact binary coalescence” 2020

TEACHING AND ADVISING

1. COADVISING MASTER STUDENT BRUNO RANDO FORASTIER 2023-2024
2. COADVISING BACHELOR STUDENT CLÀUDIA GARCIA DIAGO 2024-2025
3. TEACHING ASSISTANT: ALGEBRA AND LINEAL GEOMETRY (12h) 2024-2025
4. TEACHING ASSISTANT: ALGEBRA AND LINEAL GEOMETRY (12h) 2023-2024
5. TEACHING ASSISTANT: ALGEBRA AND LINEAL GEOMETRY (30h) 2022-2023
6. TEACHING ASSISTANT: CALCULUS I (15h) 2021-2022

Journal referee

- Refereed for Astronomy and Astrophysics
- Refereed for Astrophysics and Space Science
- Refereed for Physical Review D

Local organising committee

Institute for Theoretical Astrophysics, 2026

- Member of the LOC for the conference “AMUSE Spring School 2026”

Scientific organising committee

Institute of Space Sciences, 2024

- Member of the SOC for the conference “Cross-Collserola PhD Meeting in Astrophysics, Cosmology and Particles 2024”

PhD representative

University of Barcelona, 2022 - 2025

- I contributed to set up a transversal predoctoral community in my local institution (~70 PhD students), and acted as a ‘PhD representative’
- Our goal was to establish a network for informal collaboration, enhance our transversal skills and knowledge and create a safe space to fight against any and all forms of harassment.
- My role as a PhD representative was to organise regular workshops on transversal skills such as public speaking, outreach tips and tricks, Git courses for first-years and career in astronomy.

1. *N*-BODY MODELLING OF THE ED-2 STREAM PROGENITOR SHOWS GAIA BH₃’s FORMATION INVOLVED DYNAMICAL INTERACTIONS
Daniel Marín Pina, Mark Gieles, Sara Rastello, Clàudia Garcia-Diago, Giuliano Iorio and Marc Ardèvol, arXiv:2604.24874 (2026), submitted to A&A
2. THE ROLE OF ACCRETION EFFICIENCY, NATAL KICKS, AND ANGULAR MOMENTUM TRANSPORT IN THE FORMATION OF THE GAIA BLACK HOLES
Michela Mapelli, Cecilia Sgalletta, Johanna Müller-Horn, Giuliano Iorio, Stefano Rinaldi, Christian Burt, **Daniel Marín Pina** and Amedeo Romagnolo, arXiv:2604.12839 (2026), submitted to A&A
3. REPOPULATING THE PAIR-INSTABILITY MASS GAP WITHOUT SUSTAINED GROWTH TO MASSIVE IMBHs: THE CASE OF 47 Tuc
Debatri Chattopadhyay, **Daniel Marín Pina**, Mark Gieles, Fabio Antonini and Fotios Fronimos Poulisis, arXiv:2604.09773 (2026), submitted to arXiv
4. THE FORMATION OF GAIA BH₃
Daniel Marín Pina, Mark Gieles, Sara Rastello and Giuliano Iorio, arXiv:2510.07021 (2025), submitted to MODEST-25 IAU Conference proceedings
5. INTERACTIONS AMONG BINARY BLACK HOLES IN STAR CLUSTERS: ECCENTRIC GRAVITATIONAL WAVE CAPTURES AND TRIPLE FORMATION
Daniel Marín Pina, Mark Gieles, Tomas Andrade and Alessandro A. Trani, A&A, 698, A229 (2025)
6. DYNAMICAL FORMATION OF GAIA BH₃ IN THE PROGENITOR GLOBULAR CLUSTER OF THE ED-2 STREAM
Daniel Marín Pina, Sara Rastello, Mark Gieles, Kyle Kremer, Laura Fitzgerald and Bruno Rando Forastier, A&A, 688, L2 (2024)
7. DEMOGRAPHICS OF THREE-BODY BINARY BLACK HOLES IN STAR CLUSTERS: IMPLICATIONS FOR GRAVITATIONAL WAVES
Daniel Marín Pina and Mark Gieles, MNRAS, 527, 8369–8381 (2024)
8. BINARY-SINGLE INTERACTIONS WITH DIFFERENT MASS RATIOS: IMPLICATIONS FOR GRAVITATIONAL WAVES FROM GLOBULAR CLUSTERS
Bruno Rando Forastier, **Daniel Marín Pina**, Mark Gieles, Simon Portegies Zwart and Fabio Antonini, A&A, 697, A118 (2025)
9. EVENT HORIZON OF A CHARGED BLACK HOLE BINARY MERGER
Daniel Marín Pina, Marta Orselli and Daniele Pica, PhRvD, 106, 084012 (2022)

10. PRECURSORY COLLAPSE IN NEUTRON STAR-BLACK HOLE MERGERS
Roberto Emparan and **Daniel Marín Pina**, PhRvD, 102, 024009 (2020)

LARGE COL- LABORATION PUBLICATIONS

- 17 refereed papers in the Gaia Collaboration
- 27 refereed papers in the Virgo Collaboration

GRANTS RECEIVED

- **FPI predoctoral grant** ($\sim 100\,000\text{€}$)
Spanish ministry of Education and Professional Formation
Barcelona, Spain
2021-2025
- **Summer research contract** ($\sim 1000\text{€}$)
Institute of Space Studies of Catalonia (IEEC)
Barcelona, Spain
Summer 2019
- **Collaboration grant** (2000€)
Spanish ministry of Education and Professional Formation
Barcelona, Spain
2019
- **Erasmus+** ($\sim 1000\text{€}$)
European Commission
Trondheim, Norway
2019

SKILLS

Languages: English (C2), Spanish (Native), Catalan (Native), Japanese (B1), Chinese (B1).
Programming: Python, Mathematica, Bash, Java, Fortran, C/C++.
Specific software: PeTar, NBODY6++, Tsunami, clusterBHBdynamics.

TALKS AND POSTERS

1. INVITED SEMINAR: “THE DYNAMICAL ORIGIN OF GAIA BH₃”
MPIA Galaxy Coffee
Max Planck Institute for Astronomy, Heidelberg, Germany
December 11, 2025
2. CONTRIBUTED TALK: “THE FORMATION OF GAIA BH₃”
The Lifecycle of Stellar Black Holes
Kavli Institute for Theoretical Physics, Santa Barbara, USA
November 18, 2025
3. SEMINAR: “THE DYNAMICAL ORIGIN OF GAIA BH₃”
STRAND Theory Meeting
University of California San Diego, San Diego, USA (online due to transportation issues)
December 6, 2025
4. CONTRIBUTED TALK: “THE DYNAMICAL ORIGIN OF GAIA BH₃”
IAU Symposium 398/MODEST-25: Compact Stars and Binaries in Dense Star Clusters
Seoul National University, Seoul, South Korea
June 15–20, 2025
5. CONTRIBUTED TALK: “THE DYNAMICAL ORIGIN OF GAIA BH₃”
IAU Symposium 398/MODEST-25: Compact Stars and Binaries in Dense Star Clusters
Seoul National University, Seoul, South Korea
June 15–20, 2025
6. CONTRIBUTED TALK: “THE DYNAMICAL ORIGIN OF GAIA BH₃”
MODEST-25SPZ: Advanced Simulations and Modeling in Astrophysics
Academiehuis Zwolle, Zwolle, the Netherlands
May 19–23, 2025
7. INVITED SEMINAR: “INTERACTIONS AMONG BINARY BLACK HOLES IN STAR CLUSTERS”
Cardiff Astro Seminar series
Cardiff University, Cardiff, United Kingdom
January 22, 2025
8. CONTRIBUTED TALK: “BBH-BBH INTERACTIONS: ECCENTRIC GRAVITATIONAL WAVE CAPTURES AND TRIPLE FORMATION”
New ideas on the origin of black hole mergers
Niels Bohr International Academy, Copenhagen, Denmark
August 12–16, 2024
9. CONTRIBUTED TALK: “WHERE DO GRAVITATIONAL WAVES WITH NON-NEGLIGIBLE ECCENTRICITY COME FROM?”
Cross-Collserola PhD Meeting in Astrophysics, Cosmology and Particles 2023
Institute of Cosmos Sciences, Barcelona, Spain
October 6, 2023
10. CONTRIBUTED TALK: “DYNAMICAL BBH FORMATION AND DISRUPTION: IMPLICATIONS FOR GRAVITATIONAL WAVES”
MODEST-23: Star clusters in the post-pandemic era
Northwestern University, CIERA, Evanston, USA
August 28-September 1, 2023

11. CONTRIBUTED TALK: “THE DEMOGRAPHICS OF DYNAMICALLY-FORMED BBHs IN STAR CLUSTERS”
Nordic Winter School on Gravitational Astrophysics
Niels Bohr International Academy, Skeikampen, Norway January 29-February 3, 2023
12. CONTRIBUTED TALK: “BLACK HOLE MERGERS FROM STAR CLUSTERS”
IEEC Forum
Institute of Space Studies of Catalonia, Barcelona, Spain June 2, 2022
13. INVITED TALK: “BLACK HOLE MERGERS FROM STAR CLUSTERS”
ICCUB Winter Meeting
Institute of Cosmos Sciences, Barcelona, Spain February 7-8, 2022
14. POSTER: “DYNAMICAL BBH FORMATION AND DISRUPTION: IMPLICATIONS FOR GRAVITATIONAL WAVES”
Two in a million: The interplay between binaries and star clusters
European Southern Observatory (ESO), ESO Garching, Germany September 11-15, 2023
15. POSTER: “THE DEMOGRAPHICS OF DYNAMICALLY-FORMED BBHs IN STAR CLUSTERS”
SEA XV Scientific Meeting
Spanish Astronomical Society, La Laguna, Spain September 5-9, 2022

SELECTED OUTREACH

1. A DANCE OF STARS AND BLACK HOLES
Outreach activity for girls
Girl’s day – Heidelberg Universität, Heidelberg, Germany April 23, 2025
2. ESPECTROGRAFIA DE FORATS NEGRES I VALIDACIONS DE RELATIVITAT GENERAL AMB GW₂₅₀₁₁₄
LVK Outreach translation (to Catalan)
LVK Collaboration, Online September 9, 2025
3. PASSEIG VIRTUAL PER LA GALÀXIA
Outreach activity for kids
Centre cívic Vil·la Urània, Barcelona, Spain May 26, 2025
4. PASSEIG VIRTUAL PER LA GALÀXIA
Outreach activity for kids
Centre cívic Vil·la Urània, Barcelona, Spain May 14, 2025
5. INTERVIEW #MOTORDELA RECERCA
Instagram interview
@ubdivulga, Barcelona, Spain April 29, 2025
6. UNIVERSITAT DE L’EXPERIÈNCIA – L’UNIVERS COM UN TOT: L’UNIVERS EN CONTÍNUA EXPANSIÓ I, II, III
Outreach lecture series for senior citizens (6h)
Universitat de Barcelona, Barcelona, Spain April 1, 8 and 22, 2025
7. TELENOTÍCIES - GRAN ALINEAMENT DELS PLANETES
TV interview
TV3 - National Catalan TV, Barcelona, Spain February 28, 2025
8. UNIVERSITAT DE L’EXPERIÈNCIA – L’UNIVERS COM UN TOT: L’UNIVERS EN CONTÍNUA EXPANSIÓ I, II, III
Outreach lecture series for senior citizens (6h)
Universitat de Barcelona, Barcelona, Spain April 2, 16 and 18, 2024
9. EXPERIMENTS DE FÍSICA 2024
Outreach lecture for high school students (8h)
Universitat de Barcelona, Barcelona, Spain January 23 and 31, 2024
10. L’UNIVERS ÉS PLE DE COLORS: L’ASTROFÍSICA A TRAVÉS DE MÚLTIPLES MIRADES
Outreach lecture series
Biblioteca Gabriel García Márquez, Barcelona, Spain September 26, 2023
11. IL·LUMINEM ELS FORATS NEGRES: INICIACIÓ A L’ASTROFÍSICA I LA GRAVITACIÓ
Outreach lecture series (12h)
Centre Cívic Casa Golferichs, Barcelona, Spain April 27-June 29, 2023

12. CONSTEL·LACIONS EN 3D
Outreach activity for kids
Biblioteca Municipal de Bellavista, Les Franqueses del Vallès, Spain February 4, 2023
13. FORATS NEGRES AMB DANI MARÍN
Radio interview
La Caravana de Ràdio Estel, Barcelona, Spain February 1, 2023
14. EXPERIMENTS DE FÍSICA 2023
Outreach lecture for high school students (8h)
Universitat de Barcelona, Barcelona, Spain January 23 and 25, 2023
15. APADRINA UN DOCTORAND ROAST BATTLE (ENCONTRES TERCER CICLE)
Outreach lecture for undergraduate students
Universitat de Barcelona, Barcelona, Spain November 15, 2022
16. CONSTEL·LACIONS EN 3D
Outreach activity for kids
15a festa de la ciència, Barcelona, Spain May 29, 2022
17. IL·LUMINEM ELS FORATS NEGRES: INICIACIÓ A L'ASTROFÍSICA I LA GRAVITACIÓ
Outreach lecture series (12h)
Centre Cívic Casa Golferichs, Barcelona, Spain April 28-June 16, 2022
18. MIREM LA LLUNA DES DE LA PLAÇA DE LA PALMERA
Astronomical observation
Pla d'Acció Comunitària de la Verneda i la Pau – Associació ÀGORA, Institut de Ciències del Cosmos de la Universitat de Barcelona, Barcelona, Spain June 10, 2022
19. DESENVOLUPAMENT DE LA CARRERA PROFESSIONAL
Round table
IEEC Forum 2022, Barcelona, Spain June 2, 2022

- CONFERENCES
1. STELLAR-MASS BLACK HOLES AT THE NEXUS OF OPTICAL, X-RAY, AND GRAVITATIONAL WAVE SURVEYS
Kavli Institute for Theoretical Physics
Santa Barbara, California, USA November 3-December 4, 2025
 2. THE LIFECYCLE OF STELLAR BLACK HOLES
Kavli Institute for Theoretical Physics
Santa Barbara, California, USA November 17-20, 2025
 3. IAU SYMPOSIUM 398/MODEST-25: COMPACT STARS AND BINARIES IN DENSE STAR CLUSTERS
Seoul National University
Seoul, South Korea June 15–20, 2025
 4. MODEST-25SPZ: ADVANCED SIMULATIONS AND MODELING IN ASTROPHYSICS
Academiehuis Zwolle
Zwolle, the Netherlands May 19–23, 2025
 5. CHALLENGES AND FUTURE PERSPECTIVES IN GRAVITATIONAL- WAVE ASTRONOMY: O4 AND BEYOND
Lorentz Center
Leiden, the Netherlands October 14–18, 2024
 6. NEW IDEAS ON THE ORIGIN OF BLACK HOLE MERGERS
Niels Bohr Institute
Copenhagen, Denmark August 12–16, 2024
 7. TWO IN A MILLION: THE INTERPLAY BETWEEN BINARIES AND STAR CLUSTERS
European Southern Observatory (ESO)
ESO Garching, Germany September 11-15, 2023
 8. MODEST-23: STAR CLUSTERS IN THE POST-PANDEMIC ERA
Northwestern University, CIERA
Evanston, USA August 28-September 1, 2023

9. XV SCIENTIFIC MEETING
Spanish Astronomical Society (SEA)
Tenerife, Spain
September 5-9, 2022

SCHOOLS

1. AMUSE SPRING SCHOOL 2026
Institute for Theoretical Astrophysics
Heidelberg, Germany
March 23-March 27, 2026
2. NORDIC WINTER SCHOOL ON GRAVITATIONAL ASTROPHYSICS
Niels Bohr International Academy
Skeikampen, Norway
January 29-February 3, 2023
3. WORKSHOP ON GRAVITATIONAL WAVE MODELLING
University of Barcelona
Barcelona, Spain
October 10-11, 2022
4. GRAVITATIONAL WAVE OPEN DATA WORKSHOP
University of Heidelberg
Heidelberg, Germany (Online)
May 23-25, 2022
5. N-BODY SIMULATION LECTURE
University of Heidelberg
Heidelberg, Germany (Online)
September 27-29, 2021
6. SUMMER SCHOOL ON GRAVITATIONAL WAVE ASTROPHYSICS
Niels Bohr International Academy (NBIA)
Copenhagen, Denmark (Online due to COVID)
August 2-6, 2021